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CMP49412 – Intelligent Recommendation Systems
Assignment 2
Due: October 12, 2025 by 11:59PM

This assignment will focus on **item-based collaborative filtering**.

Question:

You have just been hired as a data scientist at a streaming platform. Your manager has observed that the current movie recommendation system is not personalized enough. To improve personalization, you have been tasked with implementing an item-based collaborative filtering algorithm from scratch.

You are provided with a CSV file called *rating.csv* that contains three columns:

1. **user_id**: the ID of a user,
2. **movie_id**: the ID of a movie,
3. **rating**: the rating that the user gave to the movie.

Your item-based collaborative filter should be built as follows:

1. Using the CSV file, create three dictionaries:
 - **user_to_movie**: A dictionary where each key is a user ID and the corresponding value is a list of movies that the user has rated.
 - **movie_to_user**: A dictionary where each key is a movie ID and the corresponding value is a list of users who have rated that movie.
 - **user_movie_to_rating**: A dictionary where each key is a tuple (user_id, movie_id) and the corresponding value is the rating.
2. Similarity between movies is calculated using the adjusted cosine similarity. Movies should be rated by at least 5 common users to be considered potential neighbors.

3. Prediction is done based on the following expression **using the 25 nearest movies**:

$$\hat{r}_{u^{(t)},i} = \mu_i + \frac{\sum_{j \in N_k(i)} \text{sim}(i,j)(r_{u^{(t)},j} - \mu_j)}{\sum_{j \in N_k(i)} |\text{sim}(i,j)|}$$

where,

- $\hat{r}_{u^{(t)},i}$ is the predicted rating for the target user on item i .
- μ_i is the average rating for the target item i computed from all users.
- μ_j is the average rating for item j .
- $r_{u^{(t)},j}$ is the target user's rating for item j .
- $\text{sim}(i,j)$ is the similarity between items i and j .
- $N_k(i)$ is the set of the k most similar items to i that the target user has rated.

Recommendation:

- Rank all unrated movies for **User25** based on the predicted ratings.
- Recommend the top 5 movies with the highest predicted ratings to User25.

Submission Requirements:

Submit a **Jupyter Notebook**, named as *assignment_2_lastName_firstName_yourID*, containing:

1. Your implementation of the item-based collaborative filter exactly as explained above.
2. The top 5 movie recommendations for **User25**.

Make sure that your notebook is clean and your code is well-documented.